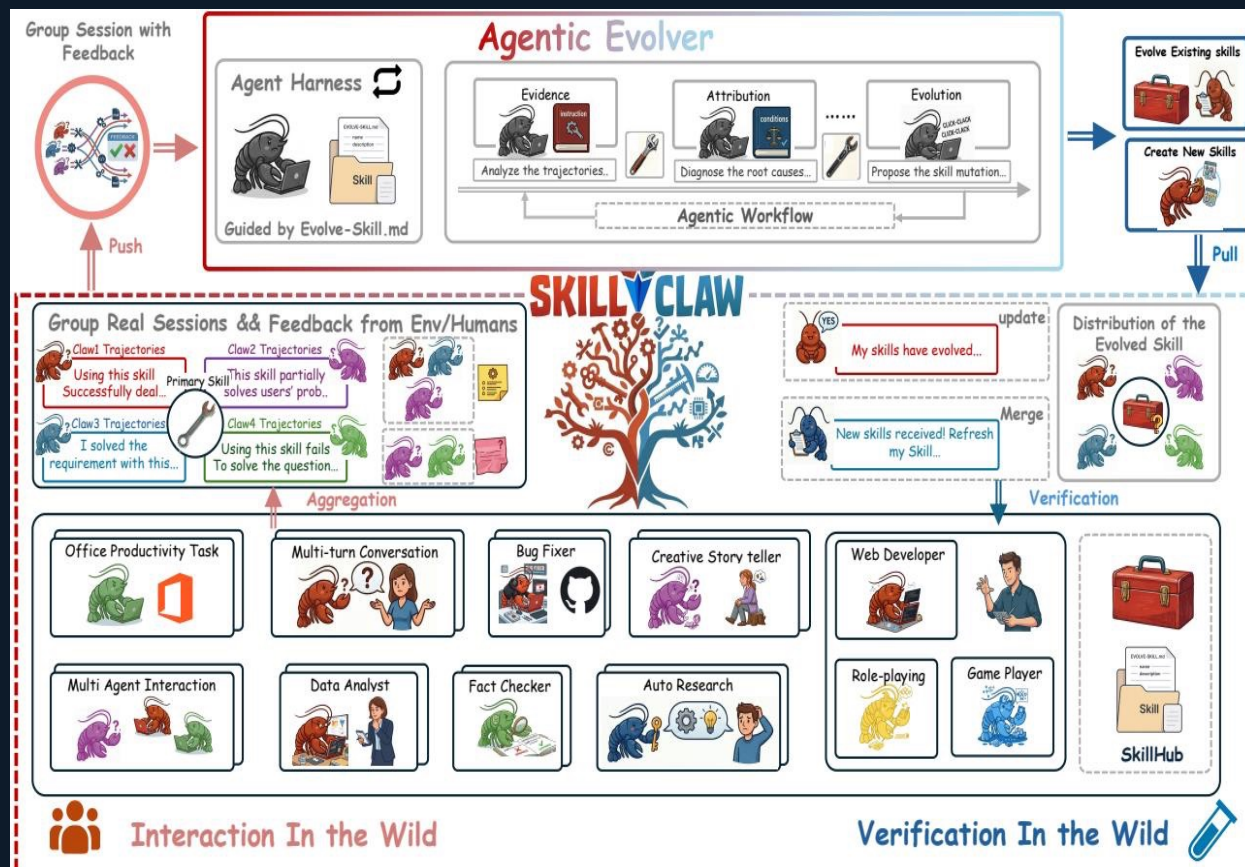




SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Closed-loop architecture enabling LLM agent skills to continuously evolve from cross-user interaction trajectories

arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently — System-Level Knowledge Never Accumulates

Static Paradigm

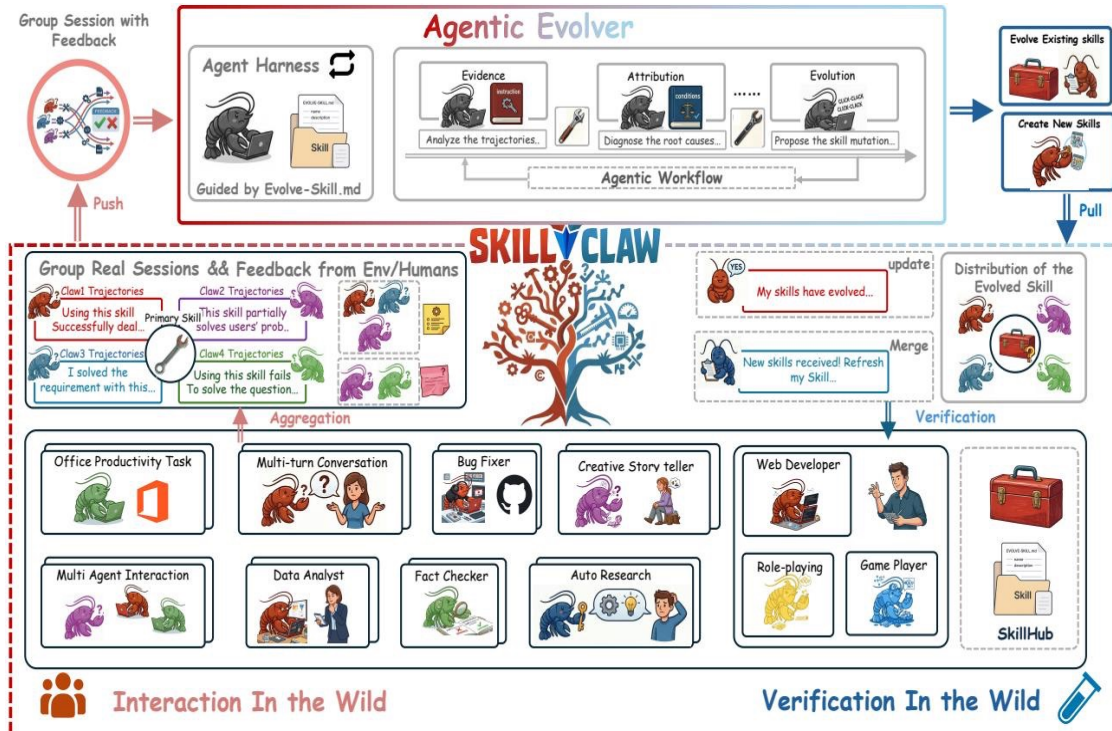
- Skills manually installed from hub
- Failures rediscovered per user
- Solutions stay in individual sessions
- Same patterns repeat across users
- No system-level improvement

SkillClaw Evolution

- Skills evolve from runtime evidence
- Failures aggregate across users
- Solutions shared in skill ecosystem
- Diverse contexts reveal boundaries
- Continuous system-level learning

Memory-based methods remain instance-tied; Skill-based methods treat libraries as static resources

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



1. Evidence

Analyze trajectories
Identify patterns

2. Attribution

Diagnose root causes
Success & failure modes

3. Evolution

Propose skill mutations
Refine or create

4. Distribution

Verify & merge
Sync to all agents

Multi-User Interaction → Session Collection → Agentic Evolver → Skill Synchronization

Agentic evolver performs open-ended reasoning over evidence — not predefined update rules

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

Stage 1: Evidence

Analyze trajectories to identify behavioral patterns across multi-user sessions

Stage 2: Attribution

Diagnose root causes of failures and successes; identify boundary conditions

Stage 3: Evolution

Propose skill mutations: refine existing skills or create new ones

Key Differentiator: Open-ended agent reasoning — not predefined rules or templates

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-Style Systems

OpenClaw • CoPaw • IronClaw • PicoClaw • ZeroClaw • NanoClaw • NemoClaw



Trajectories preserve full action–feedback causal chains → Grouped by skill → Shared evidence base

Why Cross-User Aggregation?

- Diverse contexts reveal conditions where skills work
- Different users expose different failure modes
- Single user cannot separate generalizable improvement from idiosyncratic fix
- Complementary signals needed for robust evolution

Evidence from WildClawBench evaluation using Qwen3-Max demonstrates substantial improvements

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution

1 of 6 accepted → Conservative verification

Day	Candidate Skill	Skill Function	Outcome	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & env setup	✓ Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	✗ Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	✗ Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	✗ Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	✗ Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	✗ Reject	Not admitted to next cycle

Conservative verification prevents regressions — only Day 1 candidate accepted over 6 days

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

3 of 6 accepted → Successful iterative refinement

Day	Candidate Skill	Change Summary	Outcome	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	✓ Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	✓ Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	✓ Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	✓ Accept	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	✗ Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	✗ Reject	Not admitted to next cycle

Three successful improvements before plateau — evidence-driven refinement works

Key Takeaways: Three Properties That Distinguish SkillClaw

1. Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem

2. Fully Automatic

Skill evolution driven by runtime interaction without manual curation or explicit user intervention

3. Agentic Evolution Paradigm

Skill updates produced through open-ended reasoning rather than predefined update rules

Implications: Building continuously improving agent systems requires collective evidence aggregation, conservative verification, and agentic reasoning-driven evolution pipelines